Note for plant engineers: the bearing early warning system

Status: tested on simulated data only. The numbers below show how the method behaves on a computer-made test, not on your machines. It must be re-checked on real recordings before anyone relies on it.

What it does

Every ten minutes the system listens to one second of vibration from each bearing and compares it with what that bearing sounds like when healthy. A healthy bearing is "learned" from its first days of running; no failed bearings are needed. If the vibration drifts away from the learned pattern, the system raises an alarm.

What we found

- A failing bearing was always caught eventually, but **not early**. At the setting we recommend, the alarm came on average about **7 hours** after the fault began (the early signs are smaller than normal noise).
- At the recommended setting there is about **one false alarm a day**. A false alarm means an unnecessary inspection (about Rs 2,000).
- Only about **6 in 10** of the abnormal readings were flagged at that setting. We could not reach our goal of 85 in 10 without raising roughly **33 false alarms a day**, which is far too many; you would switch it off.
- A very quiet setting (no false alarms at all) still caught every fault in the end, but only about **18 hours** after it began, when it was already well advanced.

What we recommend

Use the middle setting: about one false alarm per day, warning several hours before a fault becomes severe. Treat an alarm as "go and look at this bearing today", not as "the bearing is about to fail". A missed breakdown costs about Rs 4 lakh, a needless inspection Rs 2,000, so inspecting even a hundred alarms is cheaper than one breakdown; but too many false alarms make people ignore the system, which is why the limit of two a day matters.

What we are not sure about

- Real machines have load changes, sensor ageing and dirt that our test data does not copy. The false alarm rate could be higher.
- The test had only three failures, so the figures are rough.
- Nobody has labelled the exact moment the real bearings started to fail; the timing figures depend on that.

What we need from you

Recordings from one healthy week of each machine, and the maintenance log of past failures with dates, so we can check the system against real events.